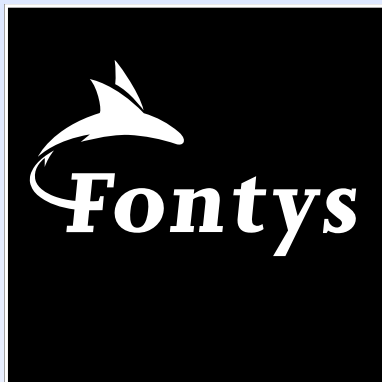




# Synthetic Data for Skin Cancer Detection

Fontys Master Applied IT

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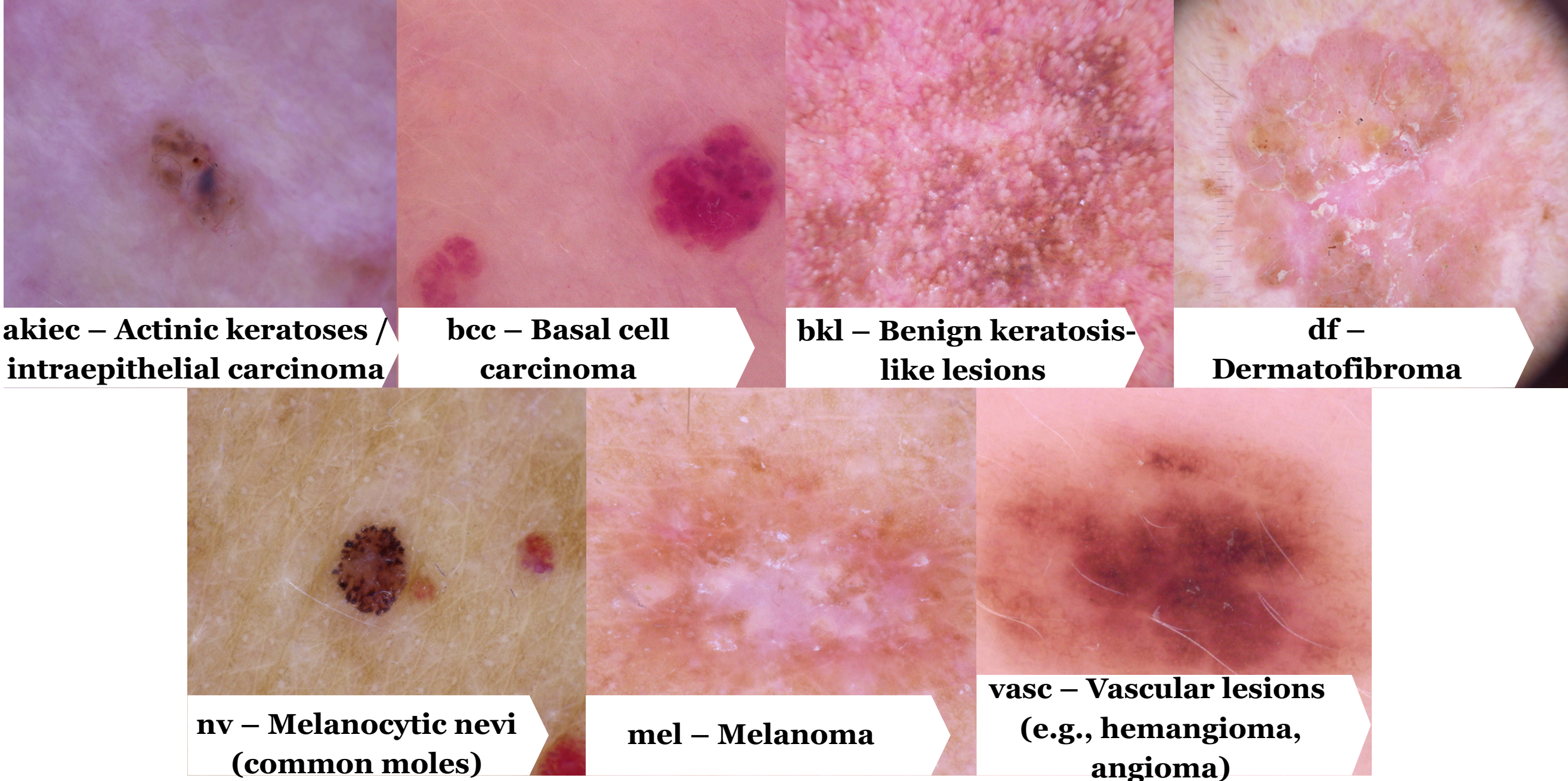
Reducing Annotation Costs, Expanding Diagnostic Access.

## PROBLEM TO SOLVE

Manual annotation of dermatological images is expensive, slow, and biased toward lighter skin tones.

## GOAL

Develop a **cost-efficient and reliable AI pipeline** that integrates **synthetic data generation** with semi-supervised **classification models** to overcome bias and annotation scarcity.



## SYNTHETIC DATA GENERATION

Approach:  
Use **GANs** and **Diffusion Models** to create realistic dermatological images conditioned on:

- **Skin tone**
- **Lesion type (melanoma, BCC, etc.)**
- **Illumination and device variability**

Key features:

- **Prompt engineering** (“melanoma on Fitzpatrick type V skin”) to expand tone diversity
- Artifact detection and **dermatologist review** to ensure **plausibility**
- **Privacy-safe** synthesis

## SEMI-SUPERVISED LEARNING (SSL) PIPELINE

Framework:  
Integrate **synthetic + real** dermatology **data** in SSL systems:

- FixMatch / MixMatch / RSMATCH
- Confidence-based **pseudo-label** filtering
- **Dynamic ratio scheduling** between real and synthetic inputs

Outcome:

- **Increases learning efficiency**
- **Reduces labeling cost**

